The Protocol Is the Result:

How Evaluation Choices Flip the Verdict on Lightweight Image Restoration

Yulin Zhao[1] (also known as Elijah Zhao)
[1]School of Economics and Law, Henan University of Economics and Law, Zhengzhou, China
20233407131@stu.huel.edu.cn ORCID: 0009-0005-4623-8465

Abstract

Lightweight image restoration is infrastructure: student-scale super-resolution and low-light enhancement models are re-implemented, compared and reported on constantly. Their reported numbers are, however, not directly comparable, because the evaluation protocol is rarely stated in full. We decompose the protocol into five operational dimensions and measure each of them on a matrix of 21 protocol cells across three tasks and five random seeds (105 evaluations in total). On the $\times 4$ super-resolution task the choice of protocol moves the reported metric by 3.889 dB—and by 19.794 dB once best-of-60 cropping is included—which is 4.98 times the 0.782 dB gain the trained model achieves over its bicubic reference. The effect is not statistical noise: seed-to-seed variation accounts for 0.49–0.93% of the protocol span on that task. Switching off the antialias flag of the downsampling operator—a single boolean, with the same weights, the same images and the same code path—moves the model's gain over bicubic from $+0.818$ dB to -1.753 dB, reversing the verdict. We also find that the claim is conditional rather than universal: on low-light enhancement the protocol span is only 0.21 times the training gain, because that task has far more headroom. Finally, a coding of 63 recent papers shows that only one states its aggregation method, none states its validation batch size, and of the 15 papers whose validation and test sets can be judged from the text, none states whether those sets coincide (the other 48 were not coded, since deciding it needs the code or the appendix). We close with six reporting requirements that would remove the ambiguity.

Keywords: evaluation protocol; reproducibility; image super-resolution; low-light enhancement; PSNR; reporting standards

1 Introduction

A student evaluates their model on the DIV2K validation split and reports 29.05 dB, against 28.23 dB for a bicubic baseline: an improvement of $+0.82$ dB. The same weights, on the same images, through the same code path, give 25.16 dB against a baseline of 26.92 dB—an improvement of -1.75 dB. The only difference is whether the downsampling operator applies antialiasing. The student did nothing wrong, but the number they report is a function of several settings that were never written down: how the metric is aggregated over images, which part of the image is evaluated, which colour channel is scored, whether the bicubic baseline is computed with antialiasing, and how many random crops are drawn before the best one is reported. Change any one of these and the same weights produce a different number. In the case we study below, they produce the opposite number.

This is not a hypothetical. We evaluate three lightweight models—a 1,224,204-parameter super-resolution network, a 69,251-parameter one, and a 564,692-parameter low-light U-Net—under eight evaluation protocols that differ only in those settings, with five random seeds per cell. The result is a matrix of 21 protocol cells, 105 evaluations in total. Across it, the protocol moves the reported metric by up to 19.794 dB, while the models themselves improve on their references by 0.78–0.86 dB on the two super-resolution tasks. The ratio between the two—the quantity that decides whether a reported improvement is a property of the method or of the measurement—reaches 4.98 on $\times 4$ super-resolution and 3.64 on $\times 2$, and falls to 0.21 on low-light enhancement.

We state the claim in its conditional form, because the low-light task is a counterexample rather than a footnote. The protocol dominates the conclusion when the available headroom and the protocol effect are of the same order. On low-light enhancement the headroom is an order of magnitude larger, the protocol effect is comparatively small, and the protocol does not dominate. Any unconditional version of the claim is refuted by our own data.

Our contributions are:

1. **A reproducible protocol audit method.** We decompose the evaluation protocol into five operational dimensions and quantify each on a matrix of 21 protocol cells $\times$ 5 seeds. The implementation groups protocols by their $(degrade, sampling)$ pair, which reduces the number of full-image forward passes from six to three per image while leaving the 60 crop evaluations of the best-of-60 protocol untouched.

2. **Evidence of a verdict reversal.** Holding the weights, the images and the code path fixed and changing only the antialias flag of the downsampling operator, the model's gain over bicubic on $\times 4$ super-resolution goes from $+0.818$ dB to -1.753 dB. The $\times 2$ task reproduces the pattern with a larger swing (3.03 dB).

3. **A coding of reporting practice.** We coded 63 pa-

pers (44 super-resolution, 19 low-light enhancement; 2019–2025) against seven reporting dimensions. One paper states its aggregation method, none states its validation batch size, and of the 15 papers whose validation and test sets can be judged from the text, none states whether those sets coincide; the remaining 48 were not coded. From these rates we derive six reporting requirements that a reader could check without rerunning anything.

Figure 1 shows the matrix. The protocol effect is visibly larger than the spread between seeds, and it is visibly not a constant: it depends on the task.

2 Related Work

2.1 Lightweight super-resolution and low-light enhancement

The super-resolution lineage relevant here begins with SRCNN [7], and continues through the residual and generator architectures of [8], [13], [12], [14] and [27]. The low-light lineage begins with Retinex decomposition and the LOL dataset [24], and continues through Retinexformer [4] and KinD [26]. The perceptual loss used by the $\times 4$ model follows [9] with a VGG feature extractor [20]; the perception–distortion trade-off that motivates it is discussed by [2]. We choose three deliberately small models (SRResNet-$\times 4$, SRCNN-$\times 2$, U-Net for low-light) not to compete with this line of work but to make the audit reproducible on a single free GPU. The question we ask is not which model is better; it is whether the numbers that distinguish models survive a change of measurement.

2.2 Protocols and reproducibility

Seven works are close neighbours to this paper, and Table 1 lists what each established and what it left open. The most directly comparable is [10], which shows that the PSNR of a set of images is not the mean of per-image PSNR, measures the difference on single image super-resolution, and gives the Jensen bound that explains it. That work isolates aggregation; we embed it in a matrix that also varies degradation, sampling, channel and boundary. [17] established that aliased versus antialiased resizing changes rankings in perceptual GAN evaluation—the narrative precedent for our reversal, though in a distributional metric rather than a distortion metric. [25] showed that evaluating real-world super-resolution on a few degradation cases drawn at random from a large degradation space yields inconsistent and at times contradictory conclusions, and [19] that protocol implementation differences prevent rankings from holding in hyperspectral super-resolution. [21] documented hidden label shifts in low-light benchmarks, which is a property of the data rather than of the protocol. [22] audited perceptual quality metrics for super-resolution and found that some correlate with human preference only weakly or negatively. [5] gave the theory of selection bias: repeatedly selecting on a validation set inflates the reported performance of the selected model.

2.3 The methodology literature

A separate line of work studies how evaluation itself distorts conclusions. [15] found that GAN improvements often arise from a larger computational budget and tuning rather than from algorithmic change, and reported no consistent advantage for any tested algorithm over the non-saturating GAN. [16] re-examined metric-learning claims and found the reported advances to be marginal at best once the experimental methodology was unified. [6] argued that test-set scores alone are insufficient, and proposed reporting the expected validation performance of the best-found model as a function of the computation budget. [3] decomposed benchmark variance into data sampling, parameter initialisation and hyperparameter choice. [18] observed that even when variance is small it is surprisingly easy to find an outlier seed that performs much better or much worse than the average.

Two distinctions matter for what follows. [3] decomposes variance on the *training* side, with weights free to vary; we hold weights and seeds fixed and vary the protocol, so the two are orthogonal rather than overlapping. And [18] concerns classification accuracy, not PSNR; the analogy we draw is mechanistic (small σ, reachable extremes), not a claim about the metric.

3 The Protocol Matrix

3.1 Five dimensions

An evaluation protocol can be specified along five operational dimensions: (i) how the per-image metric is aggregated; (ii) the validation batch size; (iii) the spatial sampling window and position; (iv) the colour channel and the boundary crop; and (v) the degradation operator and its defaults. Each can change the reported number in principle; this paper measures by how much.

One of the five has no axis in our matrix, and we say so explicitly. Dimension (ii), the validation batch size, was the loudest knob under the old aggregation: on a fixed set of 32 synthetic 96×96 crops, moving the batch from 1 to 2, 4 and 8 changed the metric by -3.56, -9.89 and -11.65 dB relative to per-block averaging, and a global all-pixel MSE dropped it by -11.85 dB. After per-image aggregation was adopted, the metric is structurally independent of how the validation set is divided into batches—a property locked in by the test suite—so batch size is no longer a variable. This is not an untested dimension; it is a dimension our fix removed. Reporting five dimensions and then varying four would otherwise read as a gap.

3.2 Eight protocols

Table 2 gives the eight protocols. They are deliberately minimal: each differs from its neighbour in one dimension, so that the effect of a dimension can be read off a pair

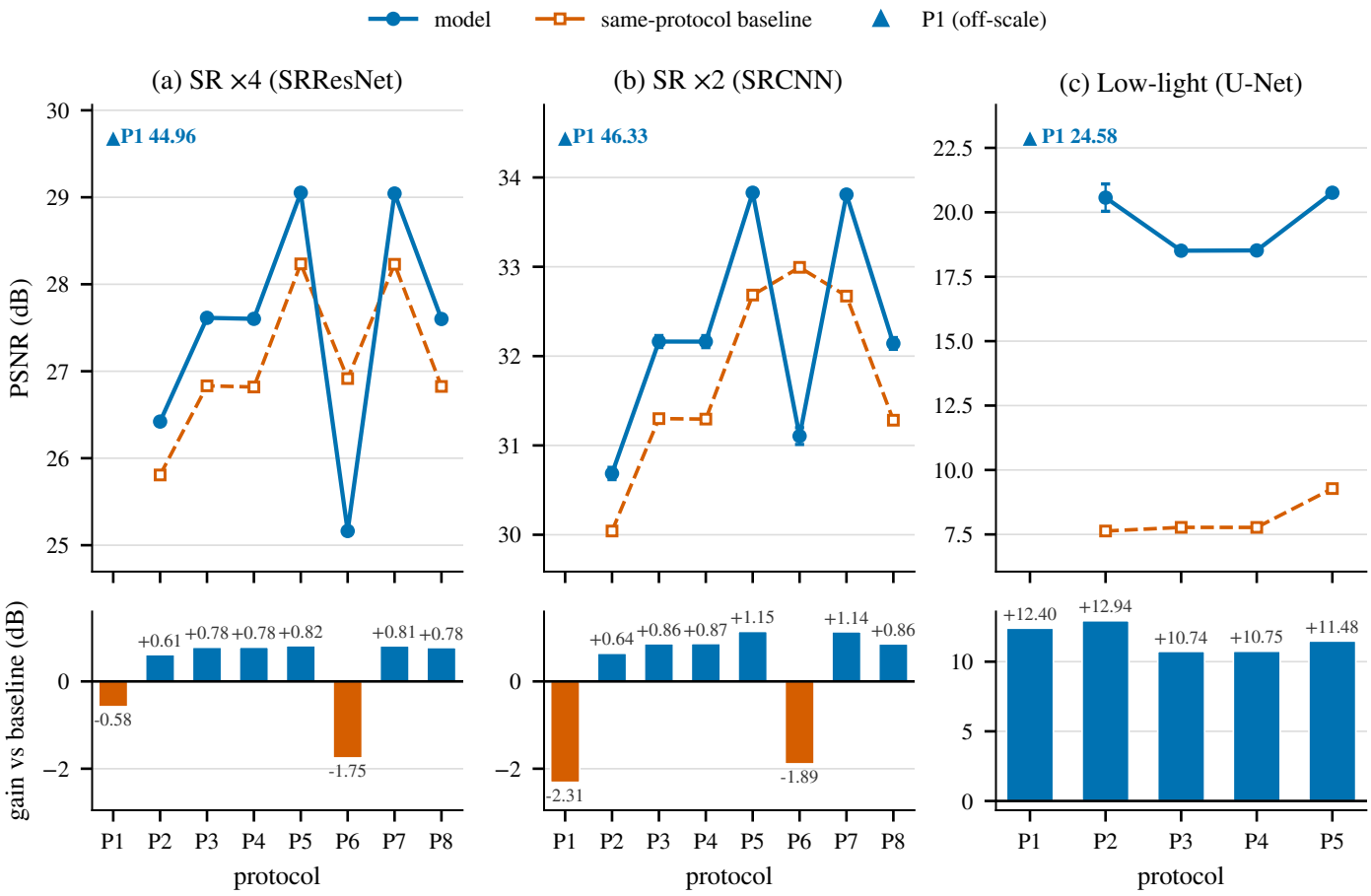


Figure 1: The protocol matrix. Each panel shows the metric for one model under eight evaluation protocols, with the five random seeds per cell summarised by a mean and its spread. The vertical distance between protocols is the protocol effect; the spread within a cell is the seed effect. The two are not of the same order, and which protocol is most demanding differs by task.

of rows rather than inferred from a regression. The most consequential pair is P5 and P6, which share every setting—full image, Y channel, 4-pixel shave—except the antialias flag of the downsampling operator. One boolean separates a positive from a negative verdict.

3.3 Grouped implementation

Protocols that share a (*degrade*, *sampling*) pair produce the same forward pass, and we exploit that. The full-image protocols P3, P4 and P5 share one forward pass per image; P6 and the P7/P8 pair share one each. Only P2 and P1 require additional passes, since they evaluate crops of the degraded image rather than the full image. The 60 crops of P1 are 60 independent forward passes, each cropping and downsampling separately; they are not cropped from a full-image output and therefore do not benefit from grouping.

On an NVIDIA T4 at 2040×1356, one full-image forward pass costs 0.2683 s and 3.15 GB for the $\times 4$ model, and 0.6624 s and 8.37 GB for the $\times 2$ model (the larger input, not the smaller model, is why memory is higher). A single 96×96 crop costs 0.0034 s. Grouping reduces the per-image cost of the $\times 4$ task from 1.8173 s to 1.0124 s, a factor of 1.80; the corresponding $\times 2$ factor is approximately 1.91, an estimate because the per-crop time was not measured separately.

4 Setup

4.1 A reproducible testbed

We use three models: SRResNet-$\times 4$ (1,224,204 trainable parameters), SRCNN-$\times 2$ (69,251) and a low-light U-Net (564,692); Table 3 collects the configuration of all three tasks in one place. The counts are measured trainable parameters; the state-dict element count is larger because it includes 2,064 batch-normalisation buffers. The $\times 4$ generator follows the residual design of [8], and the low-light network is an encoder–decoder with skip connections that predicts a residual and clamps the result.

The $\times 4$ configuration includes a perceptual loss with weight $w_{\text{percep}} = 0.006$. We report this explicitly because its presence is easy to assume and its effect easy to overstate: a paired ablation with the term switched on and off gives 27.61 dB and 27.60 dB, a difference of -0.01 dB, with a sharpness difference of 0.25%. In this weight regime the

Table 1: Nearest-neighbour work on evaluation protocols, with the part of this paper that addresses what each left open.

Work	Venue	What it established	What it left open	What this paper adds
`keles2021psnr`	PCS 2021	PSNR over a set is not the mean of per-image PSNR; gives the bound for SISR (2.07 dB measured, 2.507 dB theoretical)	Isolates aggregation only; no protocol matrix, and no seed variance	Protocol P1 (best-of-N) is the extreme case of the same effect; we quantify it against seven other protocol dimensions
`parmar2022cleanfid`	CVPR 2022	Antialiased vs aliased resizing changes FID rankings in GAN evaluation	Resize semantics only, in the perceptual/GAN metric setting	P6 vs P5 isolates antialiasing in the distortion metric; the effect is 2.72 dB, not a ranking detail
`zhang2024seal`	ICLR 2024	Random sampling of degradation cases from a large space yields inconsistent and at times contradictory conclusions	Degradation-case sampling only; channel, aggregation and boundary are held fixed	P1 vs P2 vs P3 varies crop position with the other dimensions fixed, and sets sampling beside degradation, channel and aggregation, with 5 seeds per cell
`srinivasan2026lol`	arXiv 2026	LOL benchmarks carry hidden label shifts: LOLv1 test labels render differently, and LOLv2-Real is 14% darker than the training labels	Dataset-side label shift; the evaluation protocol itself is not varied	The low-light task is our counterexample: protocol does not dominate there (ratio 0.21), so the claim is conditional
`shah2026hyperbench`	arXiv 2026	Protocol implementation differences in the Wald protocol prevent hyper-resolution rankings from holding	One task family (hyperspectral), one protocol dimension (degradation)	Three tasks and eight protocols; the protocol span exceeds the training gain on two of them
`su2025bridging`	arXiv 2025	Audits perceptual quality metrics for SR and finds they diverge from distortion metrics	Metric choice is orthogonal to the protocol dimensions this paper varies	Holds the metric fixed and varies sampling, channel, aggregation and degradation instead
`cawley2010overfitting`	JMLR 2010	Selection bias: repeated model selection on a validation set inflates the reported performance of the selected model	Gives the theory; does not measure how large the bias is under a stated protocol	Measures the same bias at 1.74 dB under P1, which is why the reporting checklist requires the selection budget N

Table 2: Protocol definitions (P1-P8). Each protocol differs from its neighbour in one dimension.

	Description	Degradation	Metric	Sampling	Aggregation
P1	Random 96x96 crops, best-of-60	torchvision bicubic, AA on	RGB	random crop	best-of-N
P2	Centre 96x96 crop (training val)	torchvision bicubic, AA on	RGB	centre crop	per-image mean
P3	Full image, RGB	torchvision bicubic, AA on	RGB	full image	per-image mean
P4	Full image, RGB + shave	torchvision bicubic, AA on	RGB + shave	full image	per-image mean
P5	Full image, Y + shave	torchvision bicubic, AA on	Y + shave	full image	per-image mean
P6	Full image, Y + shave, **no antialias**	torchvision bicubic, **AA off**	Y + shave	full image	per-image mean
P7	Full image, Y + shave, PIL bicubic	PIL bicubic	Y + shave	full image	per-image mean
P8	Full image, RGB, PIL bicubic	PIL bicubic	RGB	full image	per-image mean

(a) P5 and P6 differ in exactly one boolean: the antialias flag of the downsampling operator. Every other cell is identical, and the two rows are the paper's sign-flip experiment.
(b) The matrix covers four of the five dimensions (aggregation, sampling, metric/channel, degradation). It has no axis for validation batch size: after per-image aggregation the metric is structurally independent of the batch split, which `tests/test_protocol.py` locks in.
(c) Low-light supports P1-P5 only: it is a paired task with no synthetic downsampling, so the PIL/anti-alias rows P6-P8 do not apply.

term is inert. We do not claim that perceptual losses do not affect PSNR; SRGAN uses a 1:1 ratio, and our conclusion is confined to the 1:0.006 ratio we measured.

The repository carries 81 tests (69 in `train/tests`, 12 in `tests`; the browser end-to-end module is skipped at collection time and does not count toward that total), all

Table 3: Setup: models, data and environments.

	SR ×4	SR ×2	Low-light
Architecture	SRResNet (SRCNN-style)	SRCNN	U-Net
Trainable parameters	1,224,204	69,251	564,692
Degradation	torchvision bicubic (AA)	torchvision bicubic (AA)	none (paired)
Training images	800 (DIV2K train)	800 (DIV2K train)	485 pairs (LOL train)
Validation images	100 (DIV2K val)	100 (DIV2K val)	15 pairs (LOL eval)
Seeds	42-46	42-46	42-46
Epochs	200	100	200
Loss	Charbonnier + 0.006 x VGG (inert)	Charbonnier	Charbonnier
Environment	torch 2.11+cu128 / CUDA 12.8	torch 2.11+cu128 / CUDA 12.8	torch 2.11+cu130 / CUDA 13.0

(a) Parameter counts are measured `parameters()` values, not state-dict element counts: the ×4 model has 2,064 additional BN buffers (1,226,268 total), which are not trainable weights.
(b) The perceptual term is reported because it is in the configuration, and reported as inert because the paired ablation gives on 27.61 / off 27.60 dB (−0.01 dB, sharpness difference 0.25%).
(c) The two low-light-adjacent rows are two containers on one T4, not one: the low-light branch runs CUDA 13.0 while both SR branches run CUDA 12.8.

passing, together with three CI gates. Results are consistent to within 10^{-5} dB across the environments we tested, which is what makes the numbers below worth measuring rather than reporting.

4.2 Data, seeds and training

Training uses DIV2K train (800 images) [1] for the two super-resolution tasks and LOL train (485 pairs) [24] for low-light. Evaluation uses DIV2K val (100 images) for both super-resolution tasks and LOL eval15 (15 pairs) for low-light; low-light is paired and unsampled, at scale 1. We train five seeds (42–46) for 200 epochs, except SRCNN-×2 which trains for 100. The optimizer is Adam [11] with default betas, epsilon and no weight decay, and the learning rate follows a cosine schedule annealed to approximately zero over the full number of epochs. SSIM [23] is recorded alongside PSNR throughout, but every number reported here is PSNR.

4.3 Environments

The two super-resolution tasks run under PyTorch 2.11 with CUDA 12.8 and the low-light task under PyTorch 2.11 with CUDA 13.0; the ×2 baseline runs on CPU locally. Across environments the four reported statistics agree to 0.0 at 10^{-6} precision, against an acceptance criterion of 10^{-5} dB. CPU and GPU tensors have different SHA-256 digests but a maximum elementwise difference of 6.6×10^{-7}; we therefore do not claim bitwise reproducibility.

5 Results

5.1 Decomposing the variation

Table 4 separates three quantities: the spread across protocols, the spread across random seeds within a protocol, and the gain the trained model achieves over its reference under a standard full-image RGB protocol (P3). The seed spread must be read per task and only over the full-image protocols P3–P8, because P1 and P2 sample and therefore have a different kind of variation. On P1 the seed standard deviation is 0.2364 dB for ×4, 0.5584 dB for ×2 and 0.4148 dB for low-light; on P2 it is 0.0363 dB, 0.0685 dB and 0.5303 dB respectively. The two protocols are therefore not interchangeable either: P2 is a single fixed crop and behaves like the full-image protocols, while P1 draws 60 crops per image and does not.

The complete matrix behind every number in this section—all 21 protocol cells, with the mean over the five seeds, their standard deviation, and the minimum and maximum—is given in Table A1.

The gains over the corresponding reference, together with the mixed-basis figure that must not be used, are in Table A2.

Table 4: Protocol span versus training gain, per task.

	SR ×4	SR ×2	Low-light
Protocol span, P2-P8 (dB)	3.889	3.143	2.250
Protocol span, incl. best-of-60 P1 (dB)	19.794	15.648	6.067
Training gain under P3 (dB)	0.782	0.863	10.736
Span / training gain	4.98	3.64	0.21
Span incl. P1 / training gain	25.33	18.13	0.57
Seed sigma, full-image protocols (dB)	0.0190-0.0361	0.0268-0.0938	0.0346-0.0806
Seed sigma, P1 best-of-60 (dB)	0.2364	0.5584	0.4148
Seed sigma, P2 centre crop (dB)	0.0363	0.0685	0.5303

(a) Every quantity is same-task on both sides of every ratio. No cross-task pooled interval is given: the low-light task is a counterexample (ratio 0.21), so a pooled '5-25x' would conceal it.
(b) The low-light training gain (10.74 dB) is 4.8x its protocol span, so on that task the protocol does not dominate the conclusion. The claim is conditional: when headroom and protocol effect are of the same order, the protocol decides the conclusion.
(c) Seed sigma covers only the full-image protocols. P1 and P2 sample, so their sigma is a different quantity and is listed in its own two rows: P1 draws 60 crops per image and carries the largest spread, while P2 is a single fixed crop and behaves like the full-image protocols on the SR tasks.

The protocol span is 3.889 dB on ×4 (P5 to P6) and 19.794 dB once P1 is included, against a training gain of 0.782 dB: a ratio of 4.98, or 25.33 when P1 is counted. On

×2 the span is 3.143 dB (15.648 with P1) against a gain of 0.863 dB, a ratio of 3.64 or 18.13. On low-light the span is 2.250 dB (6.067 with P1) against a gain of 10.736 dB—a ratio of 0.21, or 0.57 with P1. The low-light ratio is a counterexample, not an error: the training gain there is 4.8 times the protocol span, so the protocol does not dominate that task. We therefore give every ratio per task and do not pool them into a single range; a pooled range would conceal the counterexample.

The seed spread is a small fraction of the protocol span on every task, but the fraction is task-dependent: 0.49–0.93% on ×4, 0.85–2.98% on ×2 and 1.54–3.58% on low-light. Each comparison pairs a task's own span with that task's own seed spread; pairing the ×4 span with the ×2 seed spread would produce a range that no measurement supports.

The protocol that is most demanding is itself task-dependent. On ×4 the lowest protocol is P6, on ×2 it is P2—so "the harshest protocol" is not a property of the protocol set.

5.2 A reversal of the verdict

Table 5 isolates the reversal. Under P5 the model exceeds the bicubic reference by +0.818 dB on ×4; under P6—the same weights, the same images, the same code path, with only the antialias flag of the downsampling operator switched off—the model trails it by −1.753 dB. The swing between the two protocols is 2.571 dB, which is 3.3 times the training gain the model is credited with. The ×2 task reproduces the pattern with a larger swing of 3.034 dB.

The mechanism differs between the two tasks, and the difference is what produces the larger ×2 swing. On ×4 both sides fall when antialiasing is disabled—the model by 3.89 dB and the baseline by 1.32 dB—leaving a scissors difference of 2.57 dB. On ×2 the model falls by 2.72 dB while the baseline *rises* by 0.31 dB, because a non-antialiased downsampling kernel is approximately the inverse of a non-antialiased upsampling kernel and the baseline cancels itself out. The sign of the reported gain is decided by the difference in how the two sides respond, and the protocol sets that difference.

Both super-resolution tasks also show a negative gain under P1, the best-of-60 protocol: −0.5793 dB on ×4 and −2.3123 dB on ×2, computed on the same 95 images on both sides. This is a definitional effect rather than a weak model. Under best-of-60 the bicubic reference reconstructs flat patches exactly: PSNR is $+\infty$ on five of the 100 images, and on those five images the model reaches at best 54.9 to 63.3 dB (×4) and 50.4 to 68.7 dB (×2). The gap is not a few decibels; it is a difference in kind, because an exact reconstruction has zero mean squared error and no network output can improve on it. The protocol manufactures a baseline that cannot be beaten. On low-light the reference is the identity map, which best-of-60 also lifts: the model sits 6.07 dB above P3 for the same weights and the same images.

On low-light the gain and the super-resolution gains are not comparable, because the references differ (identity versus bicubic). Under P3 the low-light model reaches 18.509 dB against an identity reference of 7.773 dB, a gain of +10.736 dB. We draw the protocol-domination claim here and not in the limitations, because the counterexample is the most informative part of the result: the protocol moved that task's number by 2.25 dB, and that was simply small compared with the 10.74 dB available from the model.

5.3 Reporting practice in the literature

Table 6 reports what 63 papers state. One paper (1.6%, Wilson 95% interval [0.3%, 8.5%]) states its aggregation method. No paper states its validation batch size (0/63, interval [0.0%, 5.7%]). For the 44 super-resolution papers, five give a reproducible bicubic degradation (11.4%, interval [5.0%, 24.0%]); two further papers name a Gaussian or blur kernel with parameters, which are named degradations but not the bicubic protocol, so they are excluded from that count. No super-resolution paper states its shave boundary (0/44, interval [0.0%, 8.0%]); the five low-light papers are marked not applicable, since a paired task has no synthetic downsampling and the concept does not apply, and counting them in the denominator would wrongly lower the upper bound of the interval.

Deciding whether validation and test sets coincide requires the code or the appendix, and 48 of the 63 papers were never coded for it; of the 15 that can be judged, none states it. We report this as 0 of 15 judged, not as 0 of 63, and we emphasise that a zero reporting rate is not a zero occurrence. That inference is precisely the one this paper argues against.

The channel is the one dimension on which the literature is not silent but unanimous, and the unanimity is itself the finding: 41 of the 63 papers do not state a channel, while all 22 that do state one use Y and not one states RGB. The practice is therefore consistent in the sense that nobody reports doing otherwise, and unstated in the sense that two thirds of the sample say nothing at all. We read this as the strongest available illustration of our thesis: the default is shared, so no paper looks non-conforming, and yet the default silently selects the channel that discards chroma—worth +1.45 to +2.24 dB on our own tasks. A reader comparing two numbers has no way to tell whether the chroma was scored.

The sample is drawn from arXiv and therefore over-represents open-source work; the true rate in the journal literature can only be lower. The breakdown of the degradation categories on the super-resolution subset is in Table A3; the coding decisions behind every rate reported here, including the per-paper evidence for each coding, are released with the repository.

6 Discussion

6.1 Why the protocol moves the metric so much

Aggregation is the clearest case. By Jensen's inequality, mean(PSNR) ≥ PSNR(mean MSE), because the logarithm is concave: averaging squared errors first and then taking the logarithm lowers the result. The gap is not a constant but a

Table 5: The sign reversal: identical weights, one boolean.

	SR ×4 (SRResNet)	SR ×2 (SRCNN)	Low-light (U-Net)
P5 model (dB)	29.0523	33.8290	20.7590
P5 reference (dB)	28.2344	32.6834	9.2787
P5 gain (dB)	0.8179	1.1456	11.4803
P6 model (dB)	25.1631	31.1048	–
P6 reference (dB)	26.9162	32.9937	–
P6 gain (dB)	-1.7531	-1.8889	–
Swing P5 → P6 (dB)	2.571	3.034	–
P1 gain (best-of-60, dB)	-0.5793	-2.3123	12.4027

(a) P5 and P6 share every setting but the antialias flag of the downsampling operator. The same weights, the same images and the same code path give a positive gain under P5 and a negative one under P6. Low-light has no P6, so its column shows the P5 row only.
(b) P1's negative gain on SR is a definitional effect, not a weak model: under best-of-60 the bicubic reference reconstructs flat blocks exactly (PSNR = +inf on 5 images), so its aggregate is computed on the 95-image intersection with the model. No network output can beat an exact reconstruction. Low-light's P1 gain is positive (+12.40 dB) because its reference is the identity map, which best-of-60 also lifts; the effect there is that P1 sits 6.07 dB above P3 for the same model.
(c) The ×2 swing is larger than the ×4 swing because the two sides move in opposite directions: on ×2 the model loses 2.72 dB while the baseline gains 0.31 dB.

Table 6: Reporting rates in 63 recent papers.

Reported item	Reported	Denominator	Rate	Wilson 95% CI
Aggregation method	1/63	63	1.6%	[0.3%, 8.5%]
Validation batch size	0/63	63	0.0%	[0.0%, 5.7%]
SR degradation operator reproducible	5/44	44	11.4%	[5.0%, 24.0%]
Channel NOT stated	41/63	63	65.1%	[52.8%, 75.7%]
Channel stated (all Y)	22/63	63	34.9%	[24.3%, 47.2%]
Shave boundary (SR subset only)	0/44	44	0.0%	[0.0%, 8.0%]
val / test same set	0/63	63	0.0%	[0.0%, 5.7%]
Code repository	35/63	63	55.6%	[43.3%, 67.2%]

(a) Sample: N = 63 papers (SR 44 + LLIE 19), 2019-2025, all from arXiv.
(b) Shave is counted on the SR subset only: the 5 papers marked N/A are LLIE, where a paired task has no synthetic downsampling and the concept does not apply. Reporting 0/63 would wrongly lower the upper bound of the interval.
(c) For val/test, 48 of 63 papers were never coded (deciding it needs the code or appendix), so 15 is the count of papers that can be judged: 0 of 15 state it. Zero reported is not zero occurrence – that inference is exactly what this paper criticises.
(d) The degradation-operator count is 5/44 for the bicubic protocol (4 named MATLAB implementations plus 1 explicit antialias setting). Two further papers name a Gaussian/blur kernel with parameters: named degradations, but not the bicubic protocol, so they are excluded here (7/44 under the looser rule).
(e) The two channel rows are complementary: 41/63 do not state a channel and 22/63 do, and every paper in the latter group uses Y (no paper states RGB). The rows are labelled by what was reported so that neither can be read with the opposite sign.
(f) Known bias: the sample is drawn from arXiv, so it over-represents open-source work. The true rate in the journal literature can only be lower.

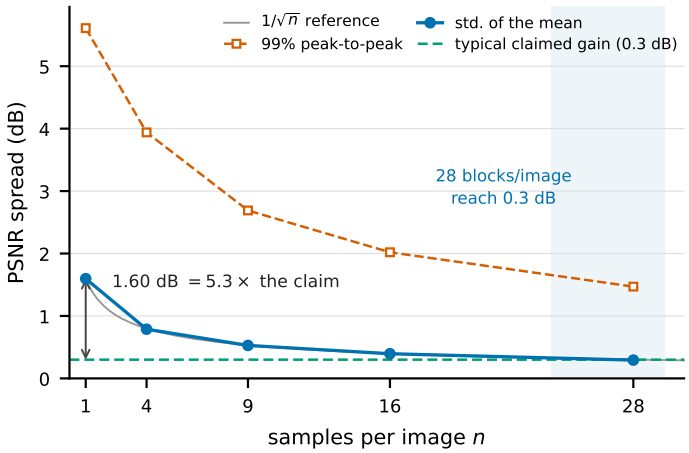


Figure 2: Noise as a function of the number of sampled crops. The metric stabilises as more crops are drawn, which is why best-of-60 protocols are not interchangeable with single-crop protocols: they report a different quantity, not a better estimate of the same one.

function of the protocol itself—1.90 dB (×4), 2.20 dB (×2) and 1.51 dB (low-light) for full-image evaluation, widening to 3.14, 3.47 and 2.77 dB when the evaluation shrinks to a single crop, because the between-image variance is then replaced by between-crop variance. This reproduces the 2.07 dB that [10] measured on single-image super-resolution; our two scales fall on either side of it.

Sampling is the second case, and it rests on two quantities we can recompute. Across the 100 DIV2K validation images the per-image PSNR spans 23.10 dB. Within a single image, the 99th percentile of the peak-to-peak spread over 28 blocks is 5.61 dB. A single 96 × 96 crop therefore carries about 1.60 dB of noise (Figure 2). The direction of the centre-crop bias reverses with the task: on super-resolution the

centre crop reads 1.2–1.5 dB low, and on low-light it reads 2.06 dB high on average—but that average should not be read as a direction. Only 9 of the 15 low-light images are higher, 6 are lower, and a single image contributes 31% of the mean. On the super-resolution side the cause is measurable: the gradient magnitude of the centre 96×96 crop is 1.37 times that of the full image, 69 of 100 images have a more textured centre, the Spearman correlation between the gradient ratio and the P2−P3 difference is -0.92, and the two sets of 69 images overlap in 64 cases. We do not claim the corresponding low-light explanation, because the original LOL images are not available locally and the hypothesis that dark scenes have brighter centres is unverified.

The degradation operator is the third case, and antialiasing is the sharpest illustration of the point. Switching the interpolation library (from torchvision to PIL, both with antialiasing) moves the metric by only 0.0090–0.0215 dB, while switching the antialias flag alone moves it by 2.7242–3.8892 dB. Paired within columns, that is a factor of 127 to 429. The smaller effect is the one that would invite a complaint about library choice; the larger one is a boolean that papers routinely leave unstated.

Channel selection raises PSNR systematically by +1.45 dB (×4), +1.67 dB (×2) and +2.24 dB (low-light), because the Y channel discards chroma. The low-light task is the most sensitive, which is consistent with real dark images carrying much larger chroma distortion than synthetic bicubic degradation.

6.2 A minimum reporting checklist

Table 7 turns the rates above into six requirements. They are deliberately limited to what a reader can check without rerunning anything: the aggregation method; the validation set and whether it is also the test set; the sampling window and position; the channel and shave boundary; the degradation operator with its antialias setting stated explicitly; and the number of seeds together with the rule used to select the best of N. The second requirement covers the validation/test overlap question, and the fifth covers both the operator and the seed count, because both answer the same question—which choices produced this number.

6.3 What this implies for reviewing

None of this asks authors to redo their experiments. It asks for one line: the aggregation method, the evaluation window, the channel, the operator with its antialias setting, the seed count, and the selection rule. A reviewer who cannot reconstruct the protocol cannot tell whether a reported gain of 0.1 dB is a property of the method or a property of the measurement, and the comparison we report here shows that the difference between the two can be several decibels.

7 Limitations

We do not claim state-of-the-art performance. Under the RGB protocol used here, KinD reports 20.87 dB on LOL (20.8665 in its Table IV; as reported, not re-run) against our 18.51 dB under P3, a difference of 2.36 dB. It is worth noting that even this comparison is protocol-governed: on the same data, the input baseline reads 7.77 dB under "RGB original" and 9.30 dB under "YCbCr 256×256", a difference of 1.53 dB from the convention alone.

The scope is two task families and three models; whether larger models reverse in the same way was not tested. All degradation is synthetic (bicubic), so blind restoration is not covered. The validation sets are small: 100 DIV2K images and 15 LOL pairs. On low-light the per-image PSNR spans 12.49–23.72 dB and the 95% interval half-width of the mean is ±2.04 dB, against ±0.87 dB for the 100-image super-resolution set; differences below 2 dB on low-light are therefore not resolvable, and we report direction only there.

The ×2 baselines were computed after the first version of the matrix, using the same code path and the same data. The literature sample covers arXiv versions only, with 43 of 63 venues recorded as unverified, so it carries an open-source bias. Cross-environment consistency was checked only across PyTorch 2.10/2.11 and CPU/GPU, with CUDA 12.8 for the two super-resolution tasks and CUDA 13.0 for low-light, and no third independent CUDA version. The perceptual-loss ablation covers one weight ratio. And the central claim is conditional: the protocol dominates on both super-resolution tasks (ratios 4.98 and 3.64) but not on low-light (0.21), because the headroom differs by an order of magnitude.

8 Conclusion

Return to the student. They had not made a mistake. They had simply not written down three lines of defaults, and those defaults were worth more than the improvement they were claiming. The protocol is not a footnote to a result; under the conditions we measured, it is part of the result.

Supporting Material

The three tables below hold the material behind the results. Table A1 gives the complete protocol matrix: every one of the 21 cells, with the mean over the five seeds, their standard deviation, and the minimum and maximum. Every protocol span quoted in the text is a difference between two rows of that table.

Every ratio in Table 4 divides one of those differences by the training gain, so the two tables can be read against each other: the span from A1, the gain from A2.

Table A2 gives the model and reference value for every protocol cell, the gain that follows from them, and the naive mixed-basis figure for the two protocols where the two aggregations have different image counts. That last column is printed only so that it can be excluded: on the ×4 task it reports a positive gain where the aligned figure is negative, because the 95-image baseline is compared against the 100-image model. The paper reports the aligned column throughout.

Table 7: A six-item minimum reporting checklist, with the literature rate for each item.

#	Item to report	Why it matters (measured here)	Literature rate
1	Aggregation: per-image mean or global MSE	Moves the metric by 1.51-2.20 dB (full image) and up to 3.47 dB (single crop)	1/63
2	Validation set and whether it is the test set	0/63 state it; deciding it needs code or appendix for 48 of 63	0/63
3	Sampling window and position	One 96x96 crop carries 1.60 dB of noise; 23.10 dB spread across 100 images	n/a
4	Channel and shave boundary	41/63 do not state the channel; 0/44 give the shave boundary	0/44 (shave)
5	Degradation operator and its defaults	Antialias alone flips the sign of the reported gain	5/44
6	Seed count and best-of-N selection	Best-of-60 raises the metric by 6.07 dB on low-light alone	n/a

(a) This table has two halves: the left two columns are the checklist, the right column gives the literature rate. Section 5.4 uses the rates, Section 6.2 uses the checklist; they are one table.
(b) Item 2 merges 'validation batch size' with 'validation/test overlap' – both are answers to the same question, 'what set was the number computed on?'. Item 5 merges the degradation operator with the seed count for the same reason: both are 'what choices produced this number?'

Table A1: Full protocol matrix (all 21 cells x 5 seeds).

Task	Protocol	Description	n images	Seeds	PSNR mean (dB)	SD	Min	Max
SR ×4	P1	Random 96x96 crops, best-of-60	100	5	44.9575	0.2364	44.5839	45.2077
SR ×4	P2	Centre 96x96 crop (training val)	100	5	26.4214	0.0363	26.3672	26.4595
SR ×4	P3	Full image, RGB	100	5	27.6139	0.0348	27.5637	27.6471
SR ×4	P4	Full image, RGB + shave	100	5	27.6024	0.0348	27.5522	27.6356
SR ×4	P5	Full image, Y + shave	100	5	29.0523	0.0361	29.0014	29.0878
SR ×4	P6	Full image, Y + shave, **no antialias**	100	5	25.1631	0.0190	25.1410	25.1854
SR ×4	P7	Full image, Y + shave, PIL bicubic	100	5	29.0433	0.0359	28.9927	29.0786
SR ×4	P8	Full image, RGB, PIL bicubic	100	5	27.6006	0.0345	27.5510	27.6337
SR ×2	P1	Random 96x96 crops, best-of-60	100	5	46.3340	0.5584	45.7223	46.9611
SR ×2	P2	Centre 96x96 crop (training val)	100	5	30.6859	0.0685	30.5702	30.7399
SR ×2	P3	Full image, RGB	100	5	32.1630	0.0670	32.0626	32.2318
SR ×2	P4	Full image, RGB + shave	100	5	32.1623	0.0671	32.0621	32.2311
SR ×2	P5	Full image, Y + shave	100	5	33.8290	0.0272	33.7855	33.8581
SR ×2	P6	Full image, Y + shave, **no antialias**	100	5	31.1048	0.0938	31.0229	31.2636
SR ×2	P7	Full image, Y + shave, PIL bicubic	100	5	33.8085	0.0268	33.7653	33.8371
SR ×2	P8	Full image, RGB, PIL bicubic	100	5	32.1415	0.0656	32.0426	32.2090
Low-light	P1	Random 96x96 crops, best-of-60	15	5	24.5767	0.4148	23.9722	25.0431
Low-light	P2	Centre 96x96 crop (training val)	15	5	20.5687	0.5303	19.9755	21.2620
Low-light	P3	Full image, RGB	15	5	18.5094	0.0363	18.4580	18.5450
Low-light	P4	Full image, RGB + shave	15	5	18.5195	0.0346	18.4657	18.5516
Low-light	P5	Full image, Y + shave	15	5	20.7590	0.0806	20.6579	20.8560

(a) The complete matrix: 21 protocol cells x 5 seeds = 105 evaluations. Low-light has no P6-P8, so the cell count is 21, not 8 x 3 x 5 = 120.
(b) Seed SD is the sample standard deviation (ddof=1) over the five seeds.

Table A2: Per-protocol gains and the naive mixed figure that must not be used.

Task	Protocol	Model (dB)	Reference (dB)	Gain (dB)	Aligned n	Naive mixed gain (not used)
SR ×4	P1	44.9575	44.7827	-0.5793	95	+0.1748
SR ×4	P2	26.4214	25.8076	0.6138	-	-
SR ×4	P3	27.6139	26.8323	0.7816	-	-
SR ×4	P4	27.6024	26.8189	0.7835	-	-
SR ×4	P5	29.0523	28.2344	0.8179	-	-
SR ×4	P6	25.1631	26.9162	-1.7531	-	-
SR ×4	P7	29.0433	28.2295	0.8138	-	-
SR ×4	P8	27.6006	26.8253	0.7753	-	-
SR ×2	P1	46.3340	48.1074	-2.3123	95	-1.7734
SR ×2	P2	30.6859	30.0418	0.6441	-	-
SR ×2	P3	32.1630	31.3001	0.8629	-	-
SR ×2	P4	32.1623	31.2948	0.8675	-	-
SR ×2	P5	33.8290	32.6834	1.1456	-	-
SR ×2	P6	31.1048	32.9937	-1.8889	-	-
SR ×2	P7	33.8085	32.6720	1.1365	-	-
SR ×2	P8	32.1415	31.2824	0.8591	-	-
Low-light	P1	24.5767	12.1740	12.4027	-	-
Low-light	P2	20.5687	7.6319	12.9368	-	-
Low-light	P3	18.5094	7.7733	10.7361	-	-
Low-light	P4	18.5195	7.7714	10.7481	-	-
Low-light	P5	20.7590	9.2787	11.4803	-	-

(a) The last column mixes the 95-image bicubic aggregate with the model's 100-image aggregate. It is shown only to be excluded: on SR x4 it reports +0.1748 where the aligned figure is -0.5793, i.e. the sign is masked by the mismatch. The paper reports the aligned column.
(b) P1 is aligned on 95 images because the bicubic reference is +inf on 5 of them.

Table A3 breaks the degradation coding down on the super-resolution subset. Its categories close exactly to the subset size, which is the check that the coding is internally consistent: a recoding that broke the closure would mean the categories no longer partition the subset.

Data Availability

All per-image evaluation values underlying every protocol cell reported in this work are publicly available in the accompanying repository under `results/per_image/`, together with the aggregated protocol matrix (`results/matrix_runs/`), the derived tables (`results/*.csv`) and the scripts that regenerate them. Each released file contains one row per test image with its PSNR and SSIM. Re-averaging these files reproduces every cell reported here; the repository includes `scripts/verify_public_release.py`, which performs that check, and the per-image release supports the distributional analysis reported in Section 6, including the per-image correlation between gradient ratio and protocol-induced shift. The DIV2K validation split and the LOL dataset are not redistributed here; they are publicly available from their original sources and are referenced accordingly.

Author Contributions

Yulin Zhao: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing—original draft, Writing— review and editing.

Funding

This research received no external funding.

Use of AI-Assisted Tools

A large-language-model assistant was used for writing and editing assistance: drafting and revising prose, and writing the analysis, table and figure scripts in `scripts/`. Every number reported in this paper was produced by those scripts from the raw evaluation outputs in `results/`, and the reported values were checked against an independent reimplementation that recomputes them from the released per-image CSVs without sharing code with the generating pipeline. The assistant is not listed as an author, and the author takes full responsibility for the content of the manuscript, including the parts it drafted, and has checked them for errors.

Table A3: F3 breakdown on the SR subset.

F3 sub-category (SR subset)	Papers	Share of 44
Named reproducible implementation (MATLAB bicubic)	4	9.1%
Explicit antialias setting	1	2.3%
"bicubic" named, implementation not given	12	27.3%
Other named degradation (Gaussian/blur, parameters given)	2	4.5%
No degradation mentioned	25	56.8%
Total (closes to the SR subset)	44	100.0%

(a) The sub-categories close exactly to the SR subset size: $4 + 1 + 12 + 2 + 25 = 44$. If a future recoding breaks that closure, the coding is inconsistent and the table is not publishable as-is.
(b) F7 code repositories: 35 of 63 report one; 1 further paper was not checked against its repository.
(c) F5 shave: on the SR subset, 10 papers say the boundary is unspecified and 34 were never coded (secondary field); 5 LLIE papers are not applicable.
(d) F6 val/test: 15 papers say nothing and 48 were never coded.

Conflicts of Interest

The author declares no conflict of interest.

References

[1] Eirikur Agustsson and Radu Timofte. NTIRE 2017 challenge on single image super-resolution: Dataset and study. In *IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 1122–1131, 2017.

[2] Yochai Blau and Tomer Michaeli. The perception-distortion tradeoff. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

[3] Xavier Bouthillier, Pierre Delaunay, Mirko Bronzi, Assya Trofimov, Brennan Nichyporuk, Justin Szeto, Nazanin Mohammadi Sepahvand, Edward Raff, Kanika Madan, Vikram Voleti, Samira Ebrahimi Kahou, Vincent Michalski, Tal Arbel, Chris Pal, Gaël Varoquaux, and Pascal Vincent. Accounting for variance in machine learning benchmarks. In *Conference on Machine Learning and Systems (MLSys)*, 2021.

[4] Yuanhao Cai, Hao Bian, Jing Lin, Haoqian Wang, Radu Timofte, and Yulun Zhang. Retinexformer: One-stage retinex-based transformer for low-light image enhancement. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

[5] Gavin C. Cawley and Nicola L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11:2079–2107, 2010.

[6] Jesse Dodge, Suchin Gururangan, Dallas Card, Roy Schwartz, and Noah A. Smith. Show your work: Improved reporting of experimental results. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019.

[7] Chao Dong, Chen Change Loy, Kaiming He, and Xiaoou Tang. Image super-resolution using deep convolutional networks. In *European Conference on Computer Vision (ECCV)*, 2014.

[8] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

[9] Justin Johnson, Alexandre Alahi, and Li Fei-Fei. Perceptual losses for real-time style transfer and super-resolution. In *European Conference on Computer Vision (ECCV)*, 2016.

[10] Onur Keleş, M. Akın Yılmaz, A. Murat Tekalp, Cansu Korkmaz, and Zafer Dogan. On the computation of PSNR for a set of images or video. In *Picture Coding Symposium (PCS)*, 2021.

[11] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, 2015.

[12] Wei-Sheng Lai, Jia-Bin Huang, Narendra Ahuja, and Ming-Hsuan Yang. Fast and accurate image super-resolution with deep laplacian pyramid networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2019.

[13] Christian Ledig, Lucas Theis, Ferenc Huszar, Jose Caballero, Andrew Cunningham, Alejandro Acosta, Andrew Aitken, Alykhan Tejani, Johannes Totz, Zehan Wang, and Wenzhe Shi. Photo-realistic single image super-resolution using a generative adversarial network. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.

[14] Bee Lim, Sanghyun Son, Heewon Kim, Seungjun Nah, and Kyoung Mu Lee. Enhanced deep residual networks for single image super-resolution. In *IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2017.

[15] Mario Lucic, Karol Kurach, Marcin Michalski, Sylvain Gelly, and Olivier Bousquet. Are GANs created equal? a large-scale study. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

[16] Kevin Musgrave, Serge Belongie, and Ser-Nam Lim. A metric learning reality check. In *European Conference on Computer Vision (ECCV)*, 2020.

[17] Gaurav Parmar, Richard Zhang, and Jun-Yan Zhu. On aliased resizing and surprising subtleties in GAN evaluation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[18] David Picard. Torch.manual_seed(3407) is all you need: On the influence of random seeds in deep learning architectures for computer vision, 2021. arXiv preprint.

[19] Ritik Shah and Marco F. Duarte. HyperBench: Standardizing and scaling synthetic evaluation for hyperspectral super-resolution, 2026. arXiv preprint.

[20] Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. In *International Conference on Learning Representations (ICLR)*, 2015.

[21] Adityan Thirumani Srinivasan. What does LOL measure? hidden label shifts in low-light enhancement benchmarks, 2026. arXiv preprint.

[22] Shaolin Su, Josep M. Rocafort, Danna Xue, David Serrano-Lozano, Lei Sun, and Javier Vazquez-Corral. Bridging the perception gap in image super-resolution evaluation, 2025. arXiv preprint.

[23] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004.

[24] Chen Wei, Wenjing Wang, Wenhan Yang, and Jiaying Liu. Deep retinex decomposition for low-light enhancement. In *British Machine Vision Conference (BMVC)*, 2018.

[25] Wenlong Zhang, Xiaohui Li, Xiangyu Chen, Yu Qiao, Xiao-Ming Wu, and Chao Dong. SEAL: A framework for systematic evaluation of real-world super-resolution. In *International Conference on Learning Representations (ICLR)*, 2024.

[26] Yonghua Zhang, Jiawan Zhang, and Xiaojie Guo. Kindling the darkness: A practical low-light image enhancer. In *ACM International Conference on Multimedia (ACM MM)*, 2019.

[27] Yulun Zhang, Kunpeng Li, Kai Li, Lichen Wang, Bineng Zhong, and Yun Fu. Image super-resolution using very deep residual channel attention networks. In *European Conference on Computer Vision (ECCV)*, 2018.